

HELSINKI MALMI, Finland

WMO: 029750

Lat: 60.255N

Lon: 25.043E

Elev: 57

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 98-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-0.6	-11.4	3.0	-7.0	-4.4	4.4	0.1	21.6	33.5	20.2	33.7	3.4	350	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	79.2	65.6	76.9	64.3	73.7	62.6	68.9	76.3	66.6	74.0	64.5	71.3	9.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	96.2	73.4	64.0	89.6	71.0	61.2	80.9	67.5	32.9	75.9	31.1	74.0	29.6	71.5	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
19.3	17.2	15.4	DB	-10.0	83.6	8.6	2.7	-16.2	85.5	-21.3	87.1	-26.1	88.6	-32.4	90.6	
			WB	-10.4	70.2	8.5	2.1	-16.6	71.8	-21.5	73.0	-26.3	74.2	-32.5	75.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	23.5	23.9	28.8	39.7	50.5	57.6	64.2	61.0	53.3	42.8	35.5	29.3
	DBStd	15.98	11.43	10.85	8.61	5.74	6.79	5.39	4.90	4.79	5.33	6.93	8.09	10.18
	HDD50	3946	822	731	658	311	75	5	0	1	28	236	436	643
	HDD65	8253	1287	1151	1123	758	453	231	75	140	352	689	886	1108
	CDD50	1245	0	0	0	3	89	233	440	342	126	12	0	0
	CDD65	77	0	0	0	0	2	9	50	16	0	0	0	0
	CDH74	576	0	0	0	0	16	82	381	97	0	0	0	0
	CDH80	71	0	0	0	0	0	7	59	5	0	0	0	0
	WSAvg	6.8	7.3	6.6	6.6	6.8	6.8	6.8	6.4	5.9	6.2	6.8	7.5	8.1
	PrecAvg	26.1	1.9	1.3	1.3	1.4	1.4	2.0	2.7	3.1	2.7	3.0	2.8	2.3
Precipitation	PrecMax	35.6	4.1	4.4	3.1	4.0	3.4	4.9	7.9	7.6	5.9	7.6	8.5	6.6
	PrecMin	16.6	0.3	0.1	0.1	0.1	0.1	0.5	0.2	0.4	0.5	0.6	0.2	0.3
	PrecStd	4.7	1.0	0.9	0.6	0.8	0.7	1.1	1.7	1.5	1.4	1.7	1.5	1.2
	PrecStd	4.7	1.0	0.9	0.6	0.8	0.7	1.1	1.7	1.5	1.4	1.7	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	42.5	42.6	52.2	64.0	76.6	80.4	84.5	80.8	69.9	58.8	51.4	48.1	
	0.4% MCWB	41.3	38.8	43.1	50.1	62.8	63.2	68.3	66.4	60.4	54.1	49.0	46.7	
	2% DB	39.6	38.9	46.3	58.8	71.9	76.6	81.0	77.3	66.5	55.8	48.6	44.8	
	2% MCWB	38.6	37.5	39.6	47.6	58.1	62.3	67.9	64.8	58.9	52.8	47.3	43.4	
	5% DB	37.8	37.3	42.6	55.0	68.1	73.1	78.6	74.9	64.5	55.0	47.8	42.9	
	5% MCWB	36.9	36.2	37.6	45.4	56.4	60.2	65.7	63.6	58.0	52.4	46.7	41.6	
	10% DB	37.0	35.6	39.2	51.5	64.2	69.5	75.3	71.8	62.5	53.3	46.1	41.0	
	10% MCWB	36.1	34.8	36.5	43.0	54.5	59.2	64.4	62.2	57.5	50.9	45.1	39.8	
	0.4% WB	41.2	39.1	44.0	51.0	64.8	67.5	71.2	70.9	62.3	55.8	49.4	46.5	
	0.4% MCDB	41.5	40.0	50.2	60.7	73.5	73.6	79.9	77.0	66.8	57.3	50.2	47.2	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	38.7	37.2	40.5	48.7	61.2	63.9	69.8	67.2	60.6	54.5	47.5	43.4	
	2% MCDB	39.3	38.0	44.2	56.6	67.8	72.5	77.9	73.6	64.6	56.0	48.2	44.2	
	5% WB	37.1	35.8	38.1	46.3	58.6	61.8	68.1	65.3	59.4	52.9	46.0	41.3	
	5% MCDB	37.7	36.4	40.9	52.5	65.6	70.2	75.8	71.9	63.1	54.6	46.8	42.2	
	10% WB	35.4	34.4	36.2	43.7	55.1	59.6	65.7	63.7	58.1	51.0	44.5	39.5	
	10% MCDB	36.0	35.0	38.3	49.6	63.1	67.2	73.0	69.4	61.7	52.8	45.2	40.4	
Mean Daily Temperature Range	MDBR	9.4	9.5	13.3	15.0	17.2	16.0	15.8	15.9	13.9	11.4	7.7	8.6	
	5% DB	7.1	8.0	16.7	22.6	23.0	22.2	20.7	21.3	16.7	12.3	7.2	7.1	
	MCDBR	7.0	7.2	11.2	13.1	11.8	10.5	9.5	10.8	10.8	9.5	7.2	6.8	
	5% WB	7.0	6.4	13.9	19.6	18.2	18.9	18.0	16.7	14.1	10.1	6.9	6.5	
	MCWBR	7.1	6.0	10.0	12.2	10.7	10.3	9.2	9.6	10.0	8.6	7.2	6.5	
	MCWBR	7.1	6.0	10.0	12.2	10.7	10.3	9.2	9.6	10.0	8.6	7.2	6.5	
Clear-Sky Solar Irradiance	taub	0.278	0.312	0.334	0.359	0.358	0.355	0.377	0.370	0.343	0.333	0.305	N/A	
	taud	2.138	2.191	2.248	2.348	2.417	2.475	2.436	2.453	2.518	2.394	2.214	N/A	
	Ebn at Noon	149	212	249	266	276	278	268	260	249	207	138	N/A	
	Edh at Noon	14	23	30	32	33	31	32	29	23	19	13	N/A	
All-Sky Solar Radiation	RadAvg	78	259	731	1219	1660	1724	1665	1288	785	343	96	46	
	RadStd	14	51	94	127	179	136	152	134	102	54	17	7	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	-256	N/A	N/A	N/A	N/A

Nomenclature: See separate page